Lab 4 – Thursday June 7, 2018

Question 1 (range, list, for)

The purpose of this question is to write a python program (script) that computes the squares for the integers from 1 to 10 by summing odd integers. For example 1 + 3 + 5 = 9 = 3*3, that is adding together the first 3 odd integers gives the square of 3.

Using the *list* and *range* functions create a list of odd integers from 1 to 19 named ***odds***. As the elements in the list ***odds*** will be summed in a loop you will need a variable named ***square*** to hold the summation. You will also need a variable named ***n*** for which the square will be computed. Assign 1 to to ***n***. Use a **for** loop to go thought the elements in ***odds***. For each element in ***odds*** add it to ***square*** to compute n squared. Print each value of ***n*** and ***square*** in 4 character positions separated by one space as shown in the sample run of the program. Add 1 to ***n*** each time through the loop.

A sample run of the program is shown below.

```
------------------------------------------------------------------

   n  n^2
   1    1
   2    4
   3    9
   4   16
   5   25
   6   36
   7   49
   8   64
   9   81
  10  100

Programmed by Stew Dent.
Date: Mon Jun  1 09:44:32 2018
End of processing.
```

Question 2 (lists, append, for, zip)

The purpose of this question is to write a python program (script) to compute lists of equally spaced time values and corresponding heights for a ball thrown straight up into the air. The lists of times and heights are parallel lists, they have the same number of elements and there is a relationship between elements at the same position in the two lists.

Use *input* to get the initial velocity v_0 of the ball in m/s as a float. Use *input* to get the integer number of intervals. Let the value of gravity G be 9.81 m/s/s. Compute the time duration the ball is in the air as $2*v_0/G$. Use a *for* loop and the *range* function to compute the equally spaced time values and add them to a list of time values. For each time value use equation 1 to compute the corresponding height of the ball. Add the height to the list of heights. Once all the times and heights have been added to the two lists print a heading as shown in the sample run of the program. Use a *for* loop and the *zip* function to go through the two lists and print each corresponding time and height in 10 characters positions to 7 decimal places separated by 1 space, one pair of values per line.

$$h(t) = v_0 t - 0.5 \cdot G \cdot t^2 \qquad equation1$$

A sample run of the program is shown below.

```
----------------------------------------------------------------

Enter the initial velocity in m/s: 2.5

Enter the number of intervals: 16

y(t) = v0*t - 0.5*g*t**2, where v0 is 2.50 m/s

     t              y(t)
 0.0000000      0.0000000
 0.0318552      0.0746607
 0.0637105      0.1393667
 0.0955657      0.1941179
 0.1274210      0.2389144
 0.1592762      0.2737561
 0.1911315      0.2986430
 0.2229867      0.3135751
 0.2548420      0.3185525
 0.2866972      0.3135751
 0.3185525      0.2986430
 0.3504077      0.2737561
 0.3822630      0.2389144
 0.4141182      0.1941179
 0.4459735      0.1393667
 0.4778287      0.0746607
 0.5096840      0.0000000

Programmed by Stew Dent.
Date: Mon Jun  1 10:06:14 2018
```

```
End of processing.
```